EASWARI ENGINEERING COLLEGE
(Autonomous)
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DEPARTMENT OF INFORMATION TECHNOLOGY

SMART FACTORY COLLABORATIVE WORKSPACE MONITORING SYSTEM

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Abstract

Smart factories use shared workspaces where employees and machines work together.

Traditional monitoring depends on manual supervision and disconnected tools.

The proposed system monitors the workspace automatically using IoT sensor data.

Collected data is analyzed for machine status, activity and workspace conditions.

Dashboard and alerts support safety and better coordination.



Introduction

Purpose of the proposed system

A real-time, IoT-based collaborative workspace monitoring system that gives supervisors continuous visibility and timely alerts.

Collaborative workspaces
Employees and machines work side by side.

Need for real-time visibility
Continuous view of activity and safety.

Limits of manual monitoring
Slow, limited coverage, prone to human error.

Role of IoT
Connected sensors and analytics monitor continuously.

Problem Statement

Delayed detection

Issues are noticed only after manual checks.

Limited monitoring coverage

Supervisors cannot watch every workspace at once.

Disconnected sensor information

Standalone sensors do not share data.

Lack of continuous insight

No ongoing view of trends or patterns.

Employee-machine interaction

Hard to monitor how people and machines work together safely.

Literature Survey

Author	Year	Description	Pros	Cons
Villani et al.	2018	Surveyed human-robot collaboration safety and interface design in industrial settings.	Established key safety and interaction guidelines.	Focused mainly on robot safety, less on full workspace monitoring.
Saleem et al.	2025	Reviewed external sensors (cameras, radars) for detecting humans in shared robot workspaces.	Comprehensive comparison of sensing technologies.	Sensor fusion adds system complexity and cost.
Maity et al.	2024	Proposed an IoT sensor network for real-time monitoring of machine temperature, vibration and status.	Enabled predictive maintenance and reduced downtime.	Requires reliable connectivity and regular sensor calibration.

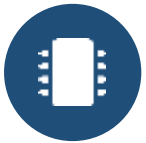
Gap addressed: this project brings workspace, machine and employee-activity monitoring together in one dashboard.

Existing System



Manual Supervision

Supervisors physically check workspace activity and safety at intervals.



Standalone Sensors

Individual sensors monitor specific machines but do not share data.



Basic CCTV

Cameras record activity but need manual review, with no real-time alerts.



Disconnected Tools

Separate monitoring tools give partial views, never one complete picture.

Most factories rely on manual supervision and disconnected tools to track workspace activity.

Existing System: Disadvantages

01

Delayed Response

Manual checks cannot catch issues as they happen.

02

Limited Coverage

Supervisors cannot monitor every workspace at once.

03

Disconnected Data

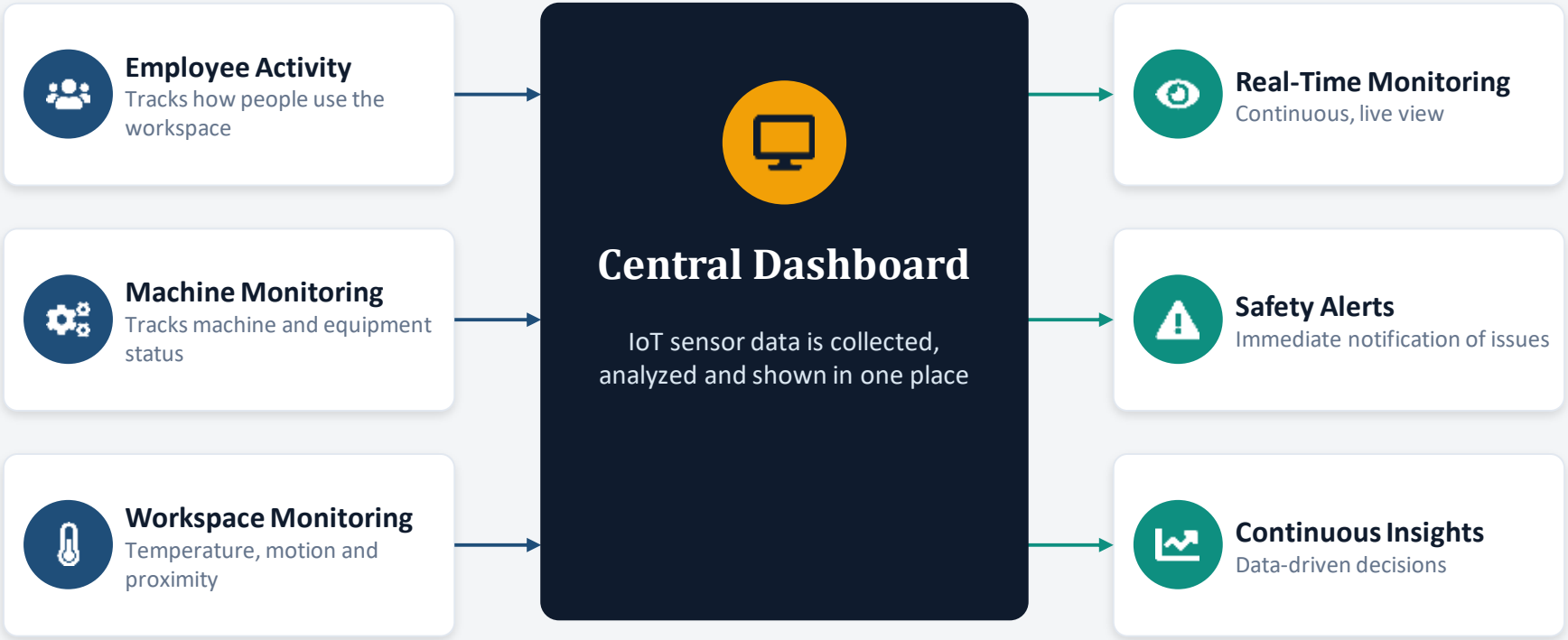
Standalone sensors do not share data for a complete picture.

04

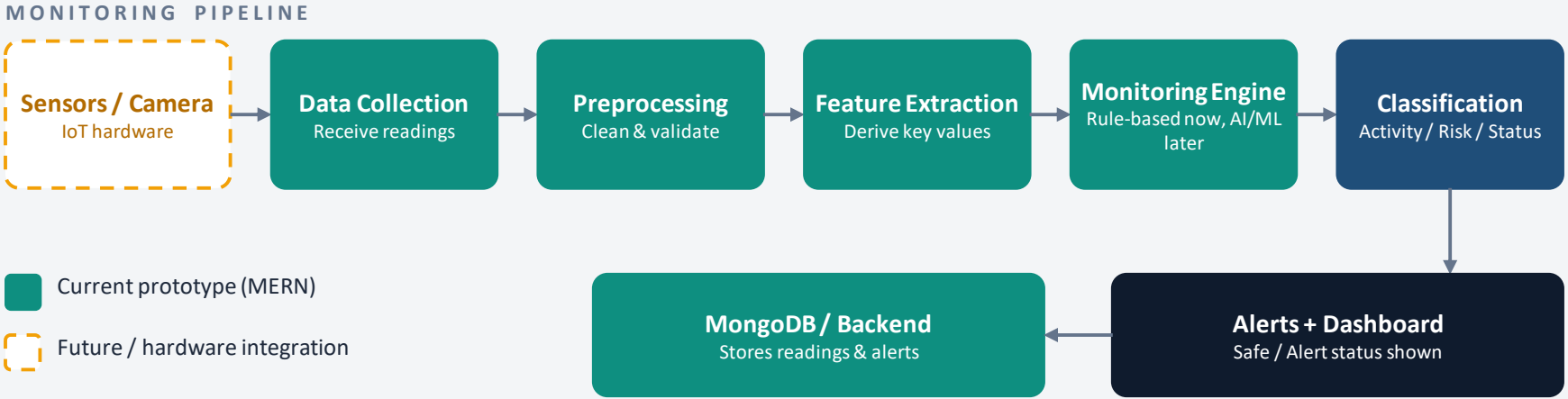
No Predictive / Continuous Insight

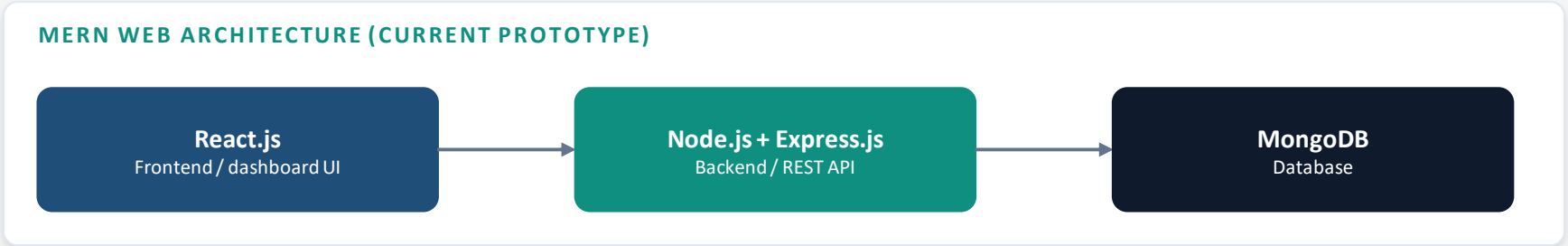
Existing tools react to problems instead of preventing them.

Proposed System



Proposed System Architecture





Modules

Authentication

Secure login and user roles

Dashboard

Central view of status and alerts

Machine Monitoring

Machine and equipment status

Sensor Monitoring

Temperature, motion, proximity readings

Workspace Monitoring

Workspace activity and safety conditions

Alert Management

Raise, view and manage alerts

Data Analytics

Historical data and trends

7 modules
built on the MERN stack

Database & Data Flow



MAJOR DATA ENTITIES (ILLUSTRATIVE DATA MODEL)

Users

name, email, role, credentials

Machines

name, location, status, last updated

Sensor Readings

sensor type, value, source, timestamp

Alerts

type, severity, source, status, time

Workspace Activity

zone, activity status, motion / proximity, time

Technology Stack

FRONTEND



React.js

HTML / CSS / JavaScript

BACKEND



Node.js

Express.js

DATABASE



MongoDB

Document database

FUTURE / INTEGRATION

- MQTT
- IoT Sensors
- AI / ML
- Computer Vision

Tools: VS Code, GitHub

System Features

Real-Time Monitoring

Live view of the workspace

Machine Status

Running, idle or fault

Sensor Readings

Temperature, motion, proximity

Safety Alerts

Warnings on unsafe conditions

Workspace Status

Zone-wise safe / alert state

Dashboard

One central view

Historical Data

Stored records to review

Analytics

Trends and summaries

Prototype note: sensor values come from sample / simulated input; real sensors are future scope.

Expected Outcome

THE PROTOTYPE IS EXPECTED TO DEMONSTRATE

- A working MERN web app with login and dashboard
- Machine, sensor and workspace data stored and displayed
- Status classification and safety alert generation
- Historical records and basic analytics
- The complete flow from data to dashboard

INTENDED BENEFITS OF THE PROPOSED SYSTEM

Higher Safety	**Real-Time Alerts**
Better Coordination	**Data-Driven Decisions**

Scope note
This is a college-level prototype. It is not deployed in a real factory, and no measured industrial results are claimed.

Future Scope

1 2 3 4

Hardware & Connectivity

- Actual IoT hardware integration
- MQTT messaging

Intelligence

- Advanced AI/ML
- Computer vision

Insight

- Predictive maintenance
- Advanced safety / risk analysis

Deployment

- Real factory deployment

References

1. Villani, V.; Pini, F.; Leali, F.; Secchi, C. Survey on human-robot collaboration in industrial settings: Safety, intuitive interfaces and applications. Mechatronics, 2018.

2. Saleem, Z.; Gustafsson, F.; Furey, E.; McAfee, M.; Huq, S. A review of external sensors for human detection in a human robot collaborative environment. Journal of Intelligent Manufacturing, 2025, 36(4), 2255-2279.

3. Li, W.; Hu, Y.; Zhou, Y.; Pham, D.T. Safe human-robot collaboration for industrial settings: A survey. Journal of Intelligent Manufacturing, 2024, 35(5), 2235-2261.

4. Maity, A.K.; Darshan; Chauhan, K. IoT-Based Industrial Equipment Monitoring System: Revolutionizing Maintenance through Smart Data Analytics. SSRN, 2024.

5. International Organization for Standardization. ISO/TS 15066:2016 Robots and Robotic Devices — Collaborative Robots, 2016.